

lab07

Caue Esteves Lambertucci

1.

```
library(RSQLite)
library(DBI)
fname <- "database.sqlite3"
conn <- dbConnect(SQLite(), fname)
```

2.

```
dbGetQuery(conn, "SELECT DISTINCT instructors.name
                  FROM instructors
                  INNER JOIN teachings ON instructors.id = teachings.instructor_id
                  INNER JOIN sections ON teachings.section_uuid = sections.uuid
                  INNER JOIN course_offerings ON sections.course_offering_uuid = course_off
                  INNER JOIN subject_memberships ON course_offerings.uuid = subject_members
                  INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
                  WHERE subjects.abbreviation = 'STAT'")
```

	name
1	MINJING TAO
2	DONALD PORTER
3	SHENG WANG
4	KUNLING HUANG
5	DONG XIA
6	JENNIFER NGUYEN
7	SEULKEE YUN
8	DUZHE WANG
9	YONGFENG WU
10	JARED BROWN
11	YUCHEN ZHOU
12	VICTOR LUO

13	GUN WOONG PARK
14	BROOK LUERS
15	DUY NGUYEN
16	JOHN GILLET
17	QUOC TRAN
18	YAOYAO XU
19	JIE ZHANG
20	LONG PHAN
21	WESLEY CHANG
22	TONG LI
23	SHANE HUBLER
24	FAN YANG
25	JU HEE CHO
26	JEEA CHOI
27	SHIXUE LIU
28	WEI-YIN LOH
29	BIN DAI
30	XIPEI YANG
31	DOUGLAS M. BATES
32	MICHAEL GEORGE ILTIS
33	LIAM JOHNSTON
34	CRYSTAL CHEN
35	YIFAN MEI
36	RUI CHEN
37	JILI WANG
38	LUWAN ZHANG
39	GINA BENNINGER
40	JIE SONG
41	XIWEN MA
42	HAODA FU
43	DEYUAN JIANG
44	JINGJIANG PENG
45	ERICA LEE DEADMAN
46	AKICHIKA OZEKI
47	MENG SONG
48	HAO TENG
49	DONGGYU KIM
50	COLIN LONGHURST
51	FREDERICK BOEHM
52	YONGSU LEE
53	GONZALO CONTADOR
54	HAO CHEN
55	XIN ZHANG

56	BO YANG
57	RUNGANG HAN
58	SEAN KENT
59	DEBRAJ DAS
60	YI LI
61	CLAIRE BOBST
62	CHAN PARK
63	SIJING LI
64	SHENGJI JIA
65	YUCHANG WU
66	ZHUANG WU
67	XIAOMAO LI
68	HYEBIN SONG
69	YOURAN QI
70	YOUNG MIN PARK
71	BIN ZHANG
72	QIURONG CUI
73	SEHO PARK
74	SHENG ZHANG
75	HUIKUN ZHANG
76	ALYSSA DIGILIO
77	YUQING XU
78	STEPHEN BERG
79	PEIGEN ZHOU
80	YAN CHEN
81	QIONG ZHANG
82	YOUNGDEOK HWANG
83	HAN CHEN
84	NICHOLAS STEPHEN KEULER
85	ZHENGXIAO WU
86	YUJIN CHUNG
87	WENWEN ZHANG
88	PERLA REYES
89	QI TANG
90	CHIA-CHIEH LIN
91	JIALE XU
92	YANG ZHAO
93	XINWEI DENG
94	MING XIE
95	JINGCI MENG
96	ISMOR FISCHER
97	JIAJIE CHEN
98	ZUOFENG SHANG

99	HAO ZHENG
100	XU XU
101	BIN ZHU
102	QUEFENG LI
103	YU QIU LIU
104	JUN LI
105	FAN GAO
106	VICTORIA MANSFIELD
107	ANQI SHI
108	KYLE HEBERT
109	XIUYU MA
110	CHENLIANG XU
111	CHANHAN HSU
112	YUANZHI LI
113	KAM-WAH TSUI
114	XINYU SONG
115	TIMOTHY IDOWU
116	XIAO NIE
117	CHEN CHENG
118	ANDREW LESLIE
119	YAOGUO XIE
120	KEVIN PACKARD
121	YI LIU
122	WENZHI CAO
123	JOSEPH DEUTSCH
124	LAN LUO
125	RYAN ZEA
126	ROBERT WARDROP
127	SANGBUM CHOI
128	XIN LI
129	SEUNGBONG HAN
130	ZHIGUO XIAO
131	RICHARD A. JOHNSON
132	RUI TANG
133	IAN BRANSTAD RILEY
134	BO HUANG
135	HEATHER MARIE BRAZEAU
136	XU HE
137	GARY HOWARD SCHROEDER
138	JUN ZHANG
139	CHENXI LI
140	SONA Z SWANSON
141	YUAN JIANG

142	JUNHEE HAN
143	LILUN DU
144	XIRAN WANG
145	ZIFENG ZHAO
146	CUIZE HAN
147	YING ZHANG
148	JOHN DAVIS
149	CHEN JING
150	JINGLAN LI
151	YI CHAI
152	THU LE
153	ALAN HUANG
154	HAOYANG FAN
155	BOWEN HU
156	TZU HSIANG HUNG
157	TUN LEE NG
158	ABIGAIL BENZINE
159	ALEXANDER COVINGTON
160	CHELSEY GREEN
161	MICHAEL KUTZLER
162	LILI ZHENG
163	REBECCA POST
164	LU YANG
165	RUOSI GUO
166	XUN ZHAO
167	MIN NIU
168	JIAN WU
169	WEI ZHENG
170	XINJIE HE
171	JIE WEI
172	CLAUDIA SOLIS LEMUS
173	XUEYAO CHEN
174	MANJUSHA KANCHARLA
175	MITCHELL PAUKNER
176	XIAOWU DAI
177	CHENGNING ZHANG
178	GINA OH
179	LUXI CAO
180	SONG WANG
181	LAM HO
182	TIEN VO
183	SHIZHEN WANG
184	SOHEIL SADEGHI

185	BINGYING XIE
186	GENG LI
187	JENNIFER BIRSTLER
188	MARIA KAMENETSKY
189	KYLE HERBET
190	JIANCHANG HU
191	TRAM TA
192	YUAN WANG
193	KAZUHIKO SHINKI
194	SIJIAN WANG
195	CHIEN-WEI CHEN
196	BEHZAD AALIPUR
197	DEREK BEAN
198	BEN ADAM HAALAND
199	ELOISA D CHAVAS
200	YALI WANG
201	VARSHA KULKARNI
202	SCOTT JOSEPH HETZEL
203	MIN JUNG LEE
204	FANG FANG
205	KRISTEN CYFFKA
206	TING-LI LIN
207	BRET LARGET
208	KATHERINE GOODE
209	ERIK NORDHEIM
210	NING FAN
211	ANRU ZHANG
212	GARVESH RASKUTTI
213	KRISHNAKUMAR BALASUBRAMANIAN
214	YAZHEN WANG
215	CHAOYANG YU
216	YILIN ZHANG
217	ZHENGJUN ZHANG
218	XIUFENG SHAO
219	YUAN LI
220	BRET HANLON
221	LILI LAN
222	CECILE ANE
223	ZHANG CHUNMING
224	LEI XU
225	MUHONG GAO
226	AUGUST JENSEN
227	XIAO GUO

228	THOMAS G. KURTZ
229	NICHOLAS HENDERSON
230	CHENSHENG KUANG
231	TAERI UHM
232	NORBERT BINKIEWICZ
233	KARL ROHE
234	JUNGWON MUN
235	JOSEP GINEBRA
236	TAO YU
237	XINXIN YU
238	HAO ZHOU
239	LIE XIONG
240	QIAN ZHIGUANG
241	SHANG WU
242	SHAN LU
243	STEPHEN AARON STANHOPE
244	LANCINE KONATE
245	GUILHERME VIEIRA NUNES LUDWIG
246	PAUL SAVARIAPPAN
247	CUICUI QI
248	KJELL DOKSUM
249	DEREK NORTON
250	KEEGAN KORTHAUER
251	KAILEI CHEN
252	THEVAASIINEN CHANDERENG
253	JOHN KANE
254	SOKOL VAKO
255	QING LI
256	YOUNG LEE
257	HYUNSEUNG KANG
258	BEHZAD AALIPUR HAFSHEJANI
259	SHULEI WANG
260	ZHONGJIE YU
261	LIN QI
262	JIWEI ZHAO
263	SUNDUZ KELES
264	SHUYUN YE
265	HUI WANG
266	KEVIN HASEGAWA ENG
267	XIAODAN WEI
268	MARIA ROJO
269	RUIFENG XU
270	YONGJOON KIM

271	NELLIE LAUGHLIN
272	MICHAEL HOGAN
273	KARL BROMAN
274	REBECCA KOSCIK
275	RONGJUN ZHU
276	BI CHENG WU
277	YUNONG LIN
278	ZHIGENG GENG
279	XIAOPING FENG
280	SHUANG HUANG
281	GUANNAN SUN
282	JEE YEON KIM
283	GREGORY SHINAULT
284	TONGHAI YANG
285	TIMO SEPPALAINEN
286	BENEDEK VALKO
287	DAVID GRIFFEATH
288	DAVID ANDERSON
289	PHILIP WOOD
290	FLORIAN BERTRAND
291	JASON RICHARD SWANSON
292	ALEXANDER KISELEV
293	SAMUEL STECHMANN
294	WAI TONG FAN
295	JUN YIN
296	STEFFEN LEMPP
297	RUIFANG SONG
298	JAMES D KUELBS
299	ANATOLE BECK
300	DONGHYUN LEE
301	ANDREJ ZLATOS
302	DANIELE CAPPELLETTI
303	MIHAELA IFRIM
304	SEBASTIEN ROCH
305	SCOTT HOTTOVY
306	JONATHON PETERSON
307	STEPHEN WAINGER
308	ALEXANDER FISH
309	GREGORIO MORENO-FLORES
310	SUKHENDU MEHROTRA
311	DIETRICH UHLENBROCK
312	LEV BORISOV
313	RICHARD A BRUALDI

314	ARNOLD MILLER
315	MATTHEW BALLARD
316	ALBRECHT KLEMM
317	PAUL M TERWILLIGER
318	STEVEN SAM
319	KEN ONO
320	ROBERT HARRON
321	YANNAN QIU
322	JOHN WILTSHIRE-GORDON
323	KYUNGMAH KIM
324	CHRISTOPHER WAGNER
325	PAUL RATHOUZ
326	STEPHEN WRIGHT
327	BENJAMIN RECHT
328	MICHAEL FERRIS
329	JESSE THOMAS HOLZER
330	HUILIN HU
331	ALBERTO DEL PIA
332	ROBERT R MEYER
333	SHI JIN
334	MARY LINDSTROM
335	ADIN-CRISTIAN ANDREI
336	MICHAEL LIOU
337	CHAOQUN MEI
338	CHRISTINA M. KENDZIORSKI
339	RONALD GANGNON
340	ROBERT WAYNE GREEN
341	COLE COOK
342	MURRAY CLAYTON
343	QI JIANG
344	JUN ZHU
345	ANNE BRUCKNER
346	TING YE
347	JUN SHAO
348	HUAIBAO FENG
349	ZIJIAN NI
350	THOMAS COOK
351	RICHARD J. CHAPPELL
352	MOO K CHUNG
353	MICHAEL NEWTON
354	BRIAN YANDELL
355	MICHELLE HARRIS
356	NORMAN DRAPER

```

357             YU MENGANG
358             MING YUAN
359             FANGFANG WANG
360             JUNHO LEE
361             GRACE WAHBA
362             JASON P FINE
363             MICHAEL RENE KOSOROK
364             YAJUAN SI
365             LU MAO
366             DAVID DEMETS
367             JAMES ANDERSON
368             EDWARD ERKER
369             SHUAI CHEN
370             MARI PALTA
371             GUANHUA CHEN
372             SRIKANTHMADHAVAN ARAVAMUTHAN
373             NATALIA DE LEON GATTI
374             GUILHERME ROSA
375             YANBING ZHENG
376             CHRISTINE SORKNESS
377             MARIAN R FISHER

```

```

dbGetQuery(conn, "SELECT COUNT(DISTINCT instructors.id) AS quantidade
                  FROM instructors
                  INNER JOIN teachings ON instructors.id = teachings.instructor_id
                  INNER JOIN sections ON teachings.section_uuid = sections.uuid
                  INNER JOIN course_offerings ON sections.course_offering_uuid = course_off
                  INNER JOIN subject_memberships ON course_offerings.uuid = subject_membersl
                  INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
                  WHERE subjects.abbreviation = 'STAT'")

```

```

quantidade
1          379

```

2.

```

dbGetQuery(conn, "SELECT COUNT(DISTINCT instructors.id) AS quantidade
                  FROM instructors
                  INNER JOIN teachings ON instructors.id = teachings.instructor_id
                  INNER JOIN sections ON teachings.section_uuid = sections.uuid
                  INNER JOIN course_offerings ON sections.course_offering_uuid = course_off

```

```
INNER JOIN subject_memberships ON course_offerings.uuid = subject_memberships.uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
WHERE subjects.abbreviation = 'STAT'")
```

quantidade
1 379

3. Professor mais difícil / mais fácil

```
dbGetQuery(conn, "SELECT instructors.name,
SUM(a_count*4 + ab_count*3.5 + b_count*3 + bc_count*2.5 + c_count*2 + d_count*1)
/ SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) AS gpa_medio
FROM instructors
INNER JOIN teachings ON instructors.id = teachings.instructor_id
INNER JOIN sections ON teachings.section_uuid = sections.uuid
INNER JOIN course_offerings ON sections.course_offering_uuid = course_offerings.uuid
INNER JOIN subject_memberships ON course_offerings.uuid = subject_memberships.uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
INNER JOIN grade_distributions ON course_offerings.uuid = grade_distributions.uuid
WHERE subjects.abbreviation = 'STAT'
GROUP BY instructors.name
ORDER BY gpa_medio ASC")
```

	name	gpa_medio
1	CHRISTINE SORKNESS	NA
2	GUILHERME ROSA	NA
3	MARIAN R FISHER	NA
4	MICHAEL RENE KOSOROK	NA
5	NATALIA DE LEON GATTI	NA
6	JAMES D KUELBS	2.607692
7	JONATHON PETERSON	2.776923
8	XIUFENG SHAO	2.778846
9	DAVID GRIFFEATH	2.805471
10	KAILEI CHEN	2.810976
11	STEPHEN WAINGER	2.811111
12	ALEXANDER FISH	2.813333
13	GINA OH	2.828883
14	MANJUSHA KANCHARLA	2.828883
15	MITCHELL PAUKNER	2.828883
16	XIAOWU DAI	2.828883
17	JUNGWON MUN	2.834604

18	SCOTT HOTTOVY	2.841379
19	ALEXANDER KISELEV	2.845133
20	HAODA FU	2.857639
21	JASON RICHARD SWANSON	2.867021
22	GREGORY SHINAULT	2.872718
23	YUJIN CHUNG	2.876682
24	KYLE HERBET	2.886986
25	MARIA KAMENETSKY	2.886986
26	BROOK LUERS	2.889455
27	BIN DAI	2.900442
28	THOMAS G. KURTZ	2.905769
29	TONGHAI YANG	2.917293
30	XIRAN WANG	2.919387
31	ZHENGXIAO WU	2.923201
32	ANATOLE BECK	2.924658
33	XIWEN MA	2.932609
34	FLORIAN BERTRAND	2.936275
35	CHENGNING ZHANG	2.940748
36	ROBERT HARRON	2.941860
37	YING ZHANG	2.946797
38	CHEN JING	2.955100
39	BO HUANG	2.956233
40	RUIFENG XU	2.959754
41	GINA BENNINGER	2.962036
42	GARY HOWARD SCHROEDER	2.963924
43	DAVID ANDERSON	2.964328
44	LIN QI	2.964416
45	BEN ADAM HAALAND	2.967114
46	ELOISA D CHAVAS	2.967114
47	MIN JUNG LEE	2.967114
48	SCOTT JOSEPH HETZEL	2.967114
49	LAM HO	2.967290
50	JENNIFER NGUYEN	2.970886
51	XIAO NIE	2.971738
52	ERICA LEE DEADMAN	2.973777
53	JUN ZHANG	2.974371
54	HEATHER MARIE BRAZEAU	2.976232
55	STEPHEN BERG	2.981969
56	PHILIP WOOD	2.982262
57	YALI WANG	2.988673
58	XU HE	2.989822
59	DONG XIA	2.992345
60	VICTOR LUO	2.992622

61	TIMO SEPPALAINEN	2.993929
62	MICHAEL FERRIS	2.995000
63	LUXI CAO	2.996523
64	MIHAELA IFRIM	2.996552
65	STEFFEN LEMPP	2.998276
66	GREGORIO MORENO-FLORES	3.000000
67	RUIFANG SONG	3.000000
68	SUKHENDU MEHROTRA	3.000000
69	IAN BRANSTAD RILEY	3.005149
70	YUANZHI LI	3.006806
71	STEPHEN WRIGHT	3.007042
72	ALBERTO DEL PIA	3.008475
73	HUILIN HU	3.008475
74	YOUNG MIN PARK	3.012450
75	SAMUEL STECHMANN	3.013043
76	YUAN JIANG	3.017171
77	TAERI UHM	3.017857
78	KAM-WAH TSUI	3.018417
79	CUIZE HAN	3.022570
80	YUAN WANG	3.022612
81	SONA Z SWANSON	3.023460
82	YUCHEN ZHOU	3.023712
83	LONG PHAN	3.024508
84	TIEN VO	3.027847
85	XINYU SONG	3.027967
86	RICHARD A BRUALDI	3.031250
87	WESLEY CHANG	3.031514
88	KUNLING HUANG	3.032448
89	SHIZHEN WANG	3.037706
90	JINGLAN LI	3.039149
91	WAI TONG FAN	3.042403
92	DUY NGUYEN	3.044562
93	NICHOLAS STEPHEN KEULER	3.045141
94	VARSHA KULKARNI	3.045924
95	ZHONGJIE YU	3.051102
96	JUNHEE HAN	3.055085
97	CHENXI LI	3.055495
98	DIETRICH UHLENBROCK	3.055556
99	LEV BORISOV	3.055556
100	BINGYING XIE	3.056132
101	SANGBUM CHOI	3.056337
102	LANCINE KONATE	3.056818
103	ARNOLD MILLER	3.057143

104	JEEA CHOI	3.057746
105	KEVIN PACKARD	3.058260
106	GENG LI	3.058872
107	HAOYANG FAN	3.059996
108	YIFAN MEI	3.062438
109	CHIEN-WEI CHEN	3.063830
110	SHIXUE LIU	3.065526
111	FANG FANG	3.065752
112	JU HEE CHO	3.067384
113	BENEDEK VALKO	3.067876
114	YONGFENG WU	3.068293
115	DOUGLAS M. BATES	3.069712
116	STEVEN SAM	3.071429
117	MICHAEL GEORGE ILTIS	3.075321
118	ISMOR FISCHER	3.077066
119	GUN WOONG PARK	3.079858
120	HUIKUN ZHANG	3.081126
121	JOHN WILTSHIRE-GORDON	3.082353
122	YONGJOON KIM	3.082367
123	SEULKEE YUN	3.083040
124	CHEN CHENG	3.084307
125	DANIELE CAPPELLETTI	3.084615
126	ANDREW LESLIE	3.085310
127	SEHO PARK	3.088114
128	RICHARD A. JOHNSON	3.088443
129	SHENG WANG	3.090047
130	YAOGUO XIE	3.090089
131	KRISHNAKUMAR BALASUBRAMANIAN	3.090741
132	ANDREJ ZLATOS	3.095238
133	DONGHYUN LEE	3.095238
134	PAUL M TERWILLIGER	3.097015
135	JINGCI MENG	3.099015
136	JIANCHANG HU	3.099393
137	YU QIU LIU	3.100315
138	DEYUAN JIANG	3.101799
139	HAO TENG	3.102080
140	ZIFENG ZHAO	3.102411
141	LILUN DU	3.103472
142	SHI JIN	3.104167
143	MIN NIU	3.106785
144	ALEXANDER COVINGTON	3.109940
145	SEBASTIEN ROCH	3.111433
146	ZUOFENG SHANG	3.115465

147	BOWEN HU	3.117920
148	KYLE HEBERT	3.118157
149	CRYSTAL CHEN	3.118306
150	ZHIGUO XIAO	3.118650
151	ALAN HUANG	3.119527
152	THU LE	3.119527
153	PERLA REYES	3.120250
154	SEUNGBONG HAN	3.120254
155	JIWEI ZHAO	3.122982
156	CHENLIANG XU	3.126014
157	JUN YIN	3.126834
158	YOUNGDEOK HWANG	3.128139
159	DONALD PORTER	3.128921
160	CHANHAN HSU	3.134167
161	TIMOTHY IDOWU	3.135092
162	TRAM TA	3.135411
163	JOHN DAVIS	3.135949
164	XIN LI	3.137337
165	JARED BROWN	3.138783
166	QIONG ZHANG	3.142224
167	BI CHENG WU	3.142322
168	RONGJUN ZHU	3.142322
169	YUNONG LIN	3.142322
170	YI LI	3.148774
171	LU YANG	3.149987
172	TING-LI LIN	3.150177
173	KRISTEN CYFFKA	3.151084
174	RUI CHEN	3.155985
175	XIPEI YANG	3.156231
176	QUOC TRAN	3.157264
177	YAZHEN WANG	3.159489
178	PEIGEN ZHOU	3.162887
179	MICHAEL HOGAN	3.164021
180	NELLIE LAUGHLIN	3.164021
181	SOHEIL SADEGHI	3.164821
182	YUCHANG WU	3.167192
183	YOURAN QI	3.167966
184	FREDERICK BOEHM	3.168681
185	SHENG ZHANG	3.169150
186	TAO YU	3.169550
187	JIE WEI	3.169589
188	XINJIE HE	3.169589
189	XUEYAO CHEN	3.169589

190	AKICHIKA OZEKI	3.170642
191	HAN CHEN	3.172547
192	MENG SONG	3.174640
193	HAO CHEN	3.174742
194	YAOTAO XU	3.174882
195	TUN LEE NG	3.177161
196	BIN ZHANG	3.179612
197	JIAN WU	3.180328
198	XUN ZHAO	3.180328
199	TONG LI	3.180372
200	FAN GAO	3.181113
201	XIN ZHANG	3.181199
202	XINWEI DENG	3.183855
203	WENWEN ZHANG	3.184142
204	SHANG WU	3.184648
205	JIALE XU	3.184762
206	MUHONG GAO	3.185088
207	HYEBIN SONG	3.185280
208	YI LIU	3.186417
209	ROBERT WARDROP	3.186426
210	WEI ZHENG	3.189254
211	KATHERINE GOODE	3.189349
212	JIE ZHANG	3.190046
213	MING XIE	3.192205
214	YILIN ZHANG	3.192470
215	AUGUST JENSEN	3.192488
216	QI TANG	3.193832
217	COLIN LONGHURST	3.195416
218	YI CHAI	3.195875
219	DEREK NORTON	3.197115
220	KEEGAN KORTHAUER	3.197115
221	SHANE HUBLER	3.197292
222	CHAN PARK	3.199041
223	CLAIRE BOBST	3.199041
224	RUNGANG HAN	3.199041
225	SEAN KENT	3.199041
226	XU XU	3.200215
227	FAN YANG	3.200251
228	MINJING TAO	3.200869
229	MARIA ROJO	3.204268
230	DONGGYU KIM	3.205146
231	JILI WANG	3.206174
232	SUNDUZ KELES	3.206349

233	YAN CHEN	3.206664
234	JOHN KANE	3.208955
235	BIN ZHU	3.209166
236	HAO ZHOU	3.211289
237	YANG ZHAO	3.211830
238	CHAOYANG YU	3.212009
239	SONG WANG	3.212232
240	CLAUDIA SOLIS LEMUS	3.213177
241	HAO ZHENG	3.213348
242	RUOSI GUO	3.214145
243	GONZALO CONTADOR	3.215695
244	RUI TANG	3.215875
245	NICHOLAS HENDERSON	3.216783
246	CHIA-CHIEH LIN	3.217348
247	DUZHE WANG	3.217694
248	JUN LI	3.219302
249	GARVESH RASKUTTI	3.219512
250	JIAJIE CHEN	3.219726
251	JIE SONG	3.221422
252	RYAN ZEA	3.223610
253	YONGSU LEE	3.226716
254	SHULEI WANG	3.227359
255	WENZHI CAO	3.229270
256	SHENGJI JIA	3.230878
257	SIJING LI	3.235288
258	CHELSEY GREEN	3.237460
259	KAZUHIKO SHINKI	3.238521
260	MICHAEL KUTZLER	3.240848
261	QUEFENG LI	3.245568
262	JINGJIANG PENG	3.246753
263	REBECCA POST	3.248538
264	ABIGAIL BENZINE	3.249023
265	ROBERT R MEYER	3.250000
266	JENNIFER BIRSTLER	3.250863
267	DEBRAJ DAS	3.252778
268	YUQING XU	3.253665
269	XIUYU MA	3.257056
270	QIURONG CUI	3.265024
271	PAUL SAVARIAPPAN	3.269175
272	ANRU ZHANG	3.271947
273	XINXIN YU	3.273276
274	SOKOL VAKO	3.274118
275	BRET HANLON	3.274387

276	BO YANG	3.276295
277	XIAO GUO	3.278666
278	HUI WANG	3.291367
279	KEVIN HASEGAWA ENG	3.291367
280	XIAODAN WEI	3.291367
281	DEREK BEAN	3.293409
282	BENJAMIN RECHT	3.297297
283	VICTORIA MANSFIELD	3.301457
284	LILI ZHENG	3.301938
285	YANNAN QIU	3.303571
286	SIJIAN WANG	3.304322
287	ZHUANG WU	3.305375
288	LUWAN ZHANG	3.305830
289	TZU HSIANG HUNG	3.306806
290	NING FAN	3.307762
291	XIAOPING FENG	3.311174
292	LIE XIONG	3.311507
293	LEI XU	3.312914
294	JOSEP GINEBRA	3.313953
295	LIAM JOHNSTON	3.315210
296	MATTHEW BALLARD	3.325581
297	KJELL DOKSUM	3.327586
298	QING LI	3.328953
299	XIAOMAO LI	3.329740
300	ANQI SHI	3.330041
301	LAN LUO	3.350858
302	SHUYUN YE	3.355795
303	YOUNG LEE	3.363533
304	KARL ROHE	3.364662
305	BEHZAD AALIPUR	3.367788
306	ALBRECHT KLEMM	3.368421
307	NORBERT BINKIEWICZ	3.370953
308	ALYSSA DIGILIO	3.386882
309	KEN ONO	3.397436
310	TING YE	3.398649
311	CHENSHENG KUANG	3.410234
312	WEI-YIN LOH	3.411338
313	CUICUI QI	3.438301
314	ERIK NORDHEIM	3.439623
315	GUILHERME VIEIRA NUNES LUDWIG	3.440221
316	ZHENGJUN ZHANG	3.440984
317	BRET LARGET	3.447608
318	HYUNSEUNG KANG	3.456140

319	CECILE ANE	3.459302
320	ROBERT WAYNE GREEN	3.460784
321	MARI PALTA	3.473684
322	SHUAI CHEN	3.473684
323	BEHZAD AALIPUR HAFSHEJANI	3.475000
324	JESSE THOMAS HOLZER	3.478261
325	JOHN GILLET	3.483620
326	QI JIANG	3.509242
327	ZHIGENG GENG	3.512411
328	COLE COOK	3.515363
329	ANNE BRUCKNER	3.516000
330	EDWARD ERKER	3.517699
331	QIAN ZHIGUANG	3.530240
332	STEPHEN AARON STANHOPE	3.532934
333	THEVAASIINEN CHANDERENG	3.541591
334	GUANNAN SUN	3.544554
335	MURRAY CLAYTON	3.545135
336	RONALD GANGNON	3.548969
337	ZHANG CHUNMING	3.563291
338	JUN ZHU	3.563474
339	JEE YEON KIM	3.571429
340	CHRISTOPHER WAGNER	3.573529
341	SHUANG HUANG	3.574468
342	JOSEPH DEUTSCH	3.590563
343	YUAN LI	3.595148
344	ZIJIAN NI	3.608696
345	JAMES ANDERSON	3.615385
346	THOMAS COOK	3.615672
347	LILI LAN	3.627445
348	CHRISTINA M. KENDZIORSKI	3.634259
349	KYUNGMAH KIM	3.638889
350	JUNHO LEE	3.642857
351	HUAIBAO FENG	3.644737
352	JUN SHAO	3.662829
353	MARY LINDSTROM	3.663004
354	NORMAN DRAPER	3.680000
355	MOO K CHUNG	3.681818
356	MICHELLE HARRIS	3.682927
357	YANBING ZHENG	3.692308
358	ADIN-CRISTIAN ANDREI	3.716463
359	KARL BROMAN	3.718391
360	JASON P FINE	3.718750
361	PAUL RATHOUZ	3.732283

362	REBECCA KOSCIK	3.746970
363	FANGFANG WANG	3.758929
364	MICHAEL NEWTON	3.761811
365	SHAN LU	3.775591
366	BRIAN YANDELL	3.785714
367	RICHARD J. CHAPPELL	3.789091
368	MICHAEL LIOU	3.813559
369	DAVID DEMETS	3.833333
370	LU MAO	3.882353
371	CHAOQUN MEI	3.886598
372	MING YUAN	3.919118
373	YU MENGANG	3.935780
374	GRACE WAHBA	3.971154
375	GUANHUA CHEN	4.000000
376	SRIKANTHMADHAVAN ARAVAMUTHAN	4.000000
377	YAJUAN SI	4.000000

3 . Disciplina mais difcil / mais facil

```
dbGetQuery(conn, "SELECT courses.name,
SUM(a_count*4 + ab_count*3.5 + b_count*3 + bc_count*2.5 + c_count*2 + d_count*1.5 + e_count*1)
/ SUM(a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) as gpa_medio
FROM courses
INNER JOIN course_offerings ON courses.uuid = course_offerings.course_uuid
INNER JOIN subject_memberships ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects ON subject_memberships.subject_code = subjects.code
INNER JOIN grade_distributions ON course_offerings.uuid = grade_distributions.course_offering_uuid
WHERE subjects.abbreviation = 'STAT'
GROUP BY courses.name
ORDER BY gpa_medio ASC")
```

	name	gpa_medio
1	Directed Study	NA
2	Practicum in Clinical Trial Data Analysis and Interpretation	NA
3	Research	NA
4	Senior Honors Thesis	NA
5	Statistical Methods for Medical Image Analysis	NA
6	Introduction to the Theory of Probability	2.886374
7	Linear Programming Methods	2.945983
8	Introduction to Probability and Mathematical Statistics I	3.026515
9	Introduction to Stochastic Processes	3.044066
10	Introduction to Probability and Mathematical Statistics II	3.057717

11	Introductory Statistics for Engineers	3.075556
12	Introduction to Statistical Methods	3.082921
13	Introduction to Combinatorics	3.102246
14	Introduction to Theory and Methods of Mathematical Statistics I	3.146679
15	Introductory Applied Statistics for Engineers	3.180488
16	Introductory Applied Statistics for the Life Sciences	3.215721
17	An Introduction to Sample Survey Theory and Methods	3.250000
18	Accelerated Introduction to Statistical Methods	3.272553
19	Applied Regression Analysis	3.326923
20	Applied Categorical Data Analysis	3.328947
21	Introduction to Theory and Methods of Mathematical Statistics II	3.334316
22	Introduction to Time Series	3.338912
23	Nonlinear Optimization I	3.420391
24	Theory of Probability I	3.426606
25	Statistical Experimental Design	3.426800
26	Intro to Biostats for Pharmacy	3.427116
27	Applied Multivariate Analysis	3.437126
28	Financial Statistics	3.468468
29	Introduction to Computational Statistics	3.468750
30	Statistical Methods for Bioscience I	3.494393
31	Statistical Methods for Bioscience II	3.498096
32	Mathematical Statistics I	3.511364
33	Statistical Methods II	3.521739
34	Statistical Methods I	3.562500
35	Theory of Probability II	3.562500
36	Mathematical Statistics	3.570720
37	Introduction to Statistical Inference	3.571429
38	Learning a Statistical Language	3.591769
39	Theory and Application of Regression and Analysis of Variance II	3.597360
40	Statistical Methods for Clinical Trials	3.611732
41	Theory and Application of Regression and Analysis of Variance I	3.622475
42	Professional Skills in Data Science	3.637931
43	Introduction to Bayesian Decision and Control I	3.666667
44	Generalized Linear Models	3.680000
45	Theory of Probability	3.693182
46	Introduction to Clinical Trials I	3.694444
47	Statistical Consulting	3.700397
48	Introduction to Biostatistics	3.713415
49	Introduction to Biostatistical Methods for Public Health	3.733333
50	Statistical Computing	3.750000
51	Statistical Methods for Spatial Data	3.752336
52	Special Topics in Statistics	3.771224
53	Applied Time Series Analysis, Forecasting and Control I	3.779412

54	Statistical Methods for Epidemiology	3.830986
55	Multivariate Analysis I	3.836634
56	Large Sample Theory of Statistical Inference	3.844828
57	Survival Analysis Theory and Methods	3.856557
58	Statistical Methods for Molecular Biology	3.875000
59	Introductory Nonparametric Statistics	3.879630
60	Nonlinear Regressn Analysis	3.909091
61	Topics in the Theory of Probability	3.941558
62	Seminar	3.951705
63	Non Parametric Statistics	3.966216
64	Statistical Model Building and Learning	3.969072
65	Estimation of Functions from Data	3.974576
66	Decision Trees for Multivariate Analysis	3.975862
67	Experimental Design I	3.984127
68	Data Science Practicum	4.000000
69	Empir Proc&Semiparmtrc Infernc	4.000000
70	Nonparametric Statistics and Machine Learning Methods	4.000000
71	Sample Survey Theory and Method	4.000000

4.

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-23 16:40:01 -03"
```